Abir Ehsan Evan

AI Developer and Research Assistant

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Professional Profile

A Computer Science Graduate passionate about Machine Learning, Data Analysis and Software Development, experienced in building LLM-powered chatbots and AI assistants for industrial and research products.

Technical Skills

Languages: Python, Java, SQL, JavaScript, HTML/CSS

Frameworks: TensorFlow Keras, PyTorch, , Flask, LangChain, MediaPipe, YOLOv5

Visualization Analysis Tools: Power BI, SPSS, Google Colab (Python-based data visualization), OpenCV

Developer Tools: Git, AWS Cloud Platform,Docker, VS Code, Visual Studio, PyCharm, PyAutoGUI, Roboflow API, Pinecone, FAISS, SQLite

Domains: Core ML (YOLOv5 training), NLP/LLM (RAG, vector embeddings), Image Processing (hand landmark detection), Web Scraping, Flask applications

Work Experience

Betopia Group

- Developed enterprise AI systems including a multimodal RAG chatbot for customer support, Automated HR orchestration and a hotel IVR concierge using LLMs, LangChain, and vector databases for real-time knowledge retrieval and automation.
- Engineered industrial-grade software solutions with OpenAI function calling and FAISS vector indexing, integrating AI assistants with SQLite databases and JSON-based systems for customer service automation and business operations.

AI Developer **January 2026 – Present**

DND Lab, University of Dhaka

- **Voice Interface Integration:** Implemented the multilingual voice-enabled system by integrating Speech-to-Text (STT) and Text-to-Speech (TTS) capabilities, involving the evaluation and deployment of various Automatic Speech Recognition (ASR) models and APIs to ensure robust performance.
- **Ethnographic Research Analysis:** Conducted on-the-ground ethnography at 3 zonal electricity offices to understand user needs and operational contexts. The insights from this HCI-oriented study directly informed the agent's design, focusing on usability and cultural relevance.

Research Assistant **October 2025 – Present**

Innovative Skills LTD, Bangladesh

- Developed a medical chatbot Web Application using NLP and AI models (OpenAI GPT / LLaMA) for symptom analysis and disease prediction.
- Built a Django REST backend to integrate AI insights with scraped data for real-time symptom checking ,doctor/hospital recommendations and appointment scheduling.

Machine Learning Intern **May 2025 - August 2025**

The Center for Astronomy, Space Science and Astrophysics (CASSA), IUB

- Analyzed cold fronts and radio minihalos in galaxy clusters using multi-wavelength data.
- Processed Chandra and MeerKAT data with Python and NumPy to support ML models and pipeline optimization.
- Cross-analyzed radio and x-ray data to validate hypotheses and contribute to publications.

Machine Learning Intern (Research) **February 2025 - October 2025**

IELTS English Language Instructor, Academia **March 2018 – July 2018**

- Conducted hands-on training and classes to improve students' English speaking, writing, and listening skills, guiding them toward language proficiency through lessons and study sessions.

Education

Independent University Bangladesh **2020–2025** ***BSc in Computer Science***
Focus on programming, computational fundamentals, and machine learning, with active involvement in research projects and extracurricular activities.

Academia **2019** *International Advanced Levels (A' Levels),*
Subjects : Physics, Mathematics.

Academia **2017** *IGCSE O' Levels,*
Subjects : Pure Mathematics, Maths-B, English Language B, Physics, Chemistry, Bangla.

Projects (Selected)

Grand Betopia AI Concierge (Automated Hotel Booking and Service System)

- Engineered an enterprise-grade AI-driven automated system leveraging RAG and OpenAI function calling to automate hotel guest inquiries and reservation management, reducing front-desk workload by handling end-to-end bookings and in-house service.
- Implemented a modular architecture with an intelligence layer (GPT-4o), knowledge layer (FAISS vector database), and action layer (function calling) that seamlessly synchronized SQLite relational data with JSON-based management exports for complete transaction processing.
- Developed key business logic features including natural language date parsing, overlap prevention for double-bookings, and dynamic pricing calculations, resulting in automated guest data extraction, structured JSON receipts, and reliable reservation management.

Automated HR Orchestration and Interview Scheduling Platform

- Built a centralized HR Automation Hub to manage the full employee lifecycle, from recruitment to inter-company interview transitions.
- Built an AI-powered pipeline using GPT-4o-mini for resume scoring, integrated Google Drive API for automated document ingestion, and utilized Python (Pandas, Openpyxl) for structured candidate reporting.

- Automated 90% of manual onboarding tasks, including one-click document generation (Offer/Warning letters) and instant platform provisioning (MS Teams/WhatsApp), resulting in 100% elimination of "ghost access" security risks through a centralized system kill-switch.

Certifications And Awards

- **Completion of Stanford University and DeepLearning.**
  AI Machine Learning Specialization on Coursera.
- Completion of workshop on Deep Learning and LLM models (Dhaka University)
- Student Researcher of Researching Endangered Languages. SCMLE, IUB.
- Academia Intra-School Debate Champion
- Edexcel High Achievers Award IGCSE